Name: Student ID:

1. Researchers took 48 resumes and randomly assigned them to 48 bank managers who were asked whether they would promote this candidate or not. The resumes were identical except the name: 24 of them had names associated with women, 24 of them had names associated with men. The researchers' results can be found in the `promote` data frame in the `stat20data` package. Did gender played a role in promotion decisions? We'll assess this question with a hypothesis test.

- Write down a mathematical expression for each of the null and alternative hypotheses.

- Write an `infer` pipeline to compute the observed test statistic. Copy your code below.

- What type of plot can be used to visualize the empirical distribution of the data? Sketch a possibility for what the data might look like if it were *consistent* with the null hypothesis. Label your axes.

- Write an `infer` pipeline to construct the null distribution of statistics and then calculate the p-value for the hypothesis test. Write the code you used below.

- Interpret the p-value. How consistent is the null hypothesis with the observed data?

2. In the 2016 NBA basketball season, Stephen Curry took 27 shots beyond 30 feet and made 48.1% of them. The `curry` data frame within the `stat20data` library contains these results. Everett would like to assess whether this data is consistent with the notion that Curry has a long-range shooting percentage that is no different from the average NBA player (7.5%), and decides to perform a hypothesis test with the `infer` library. Unfortunately, he runs into trouble! Some of the `infer` functions won't work on his RStudio session.

- Write down a mathematical expression for each of the null and alternative hypotheses.

- Write an `infer` pipeline to generate and save 500 simulations of 27 shot attempts by Curry under the null hypothesis.

- Everett tries to use the `calculate()` function to construct a null distribution with his simulations, but finds it's not working. Write `dplyr` code that will complete the calculation. Copy your code below.

- Everett then tries to use the `visualize()` function to plot a null distribution, but finds it's not working, either! Write `ggplot` code that will help Everett visualize the distribution. Copy your code below.

- Finally, Everett tries the `get_p_value()` function to calculate a p-value on his RStudio session; it's not working!! Write `dplyr` code that will help Everett calculate the p-value. Copy your code below.